

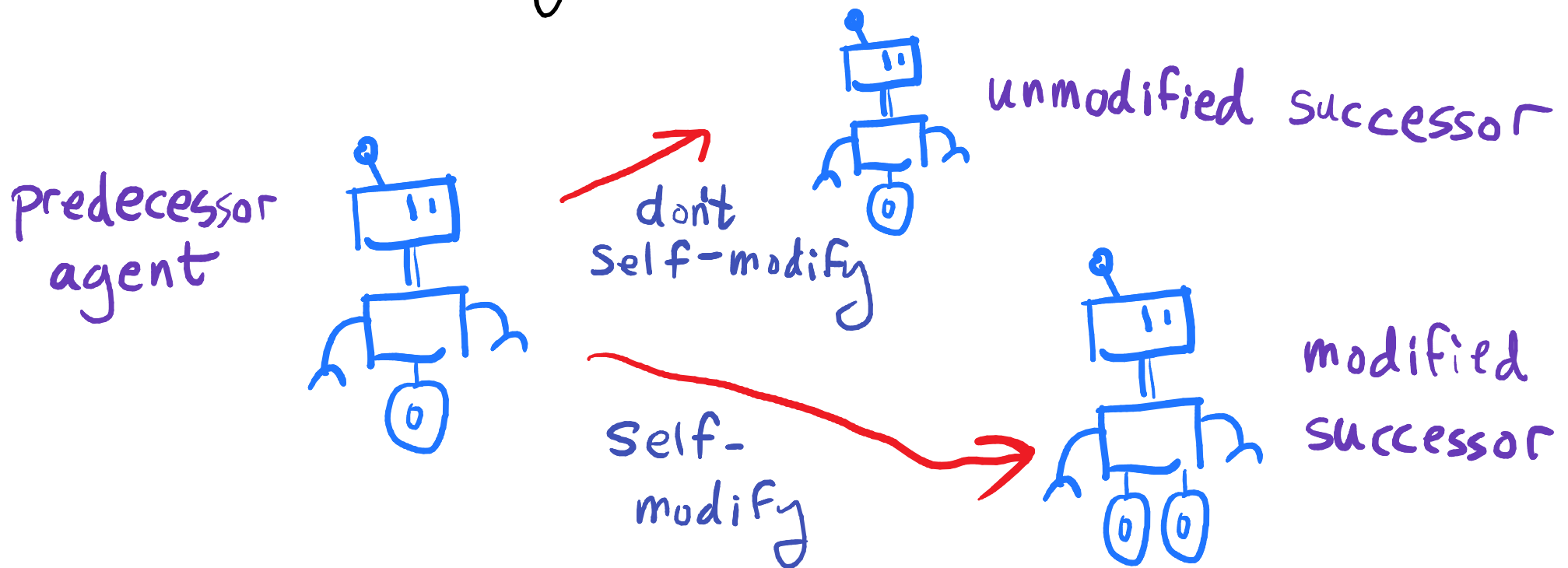
Understanding Trust

Abram Demski

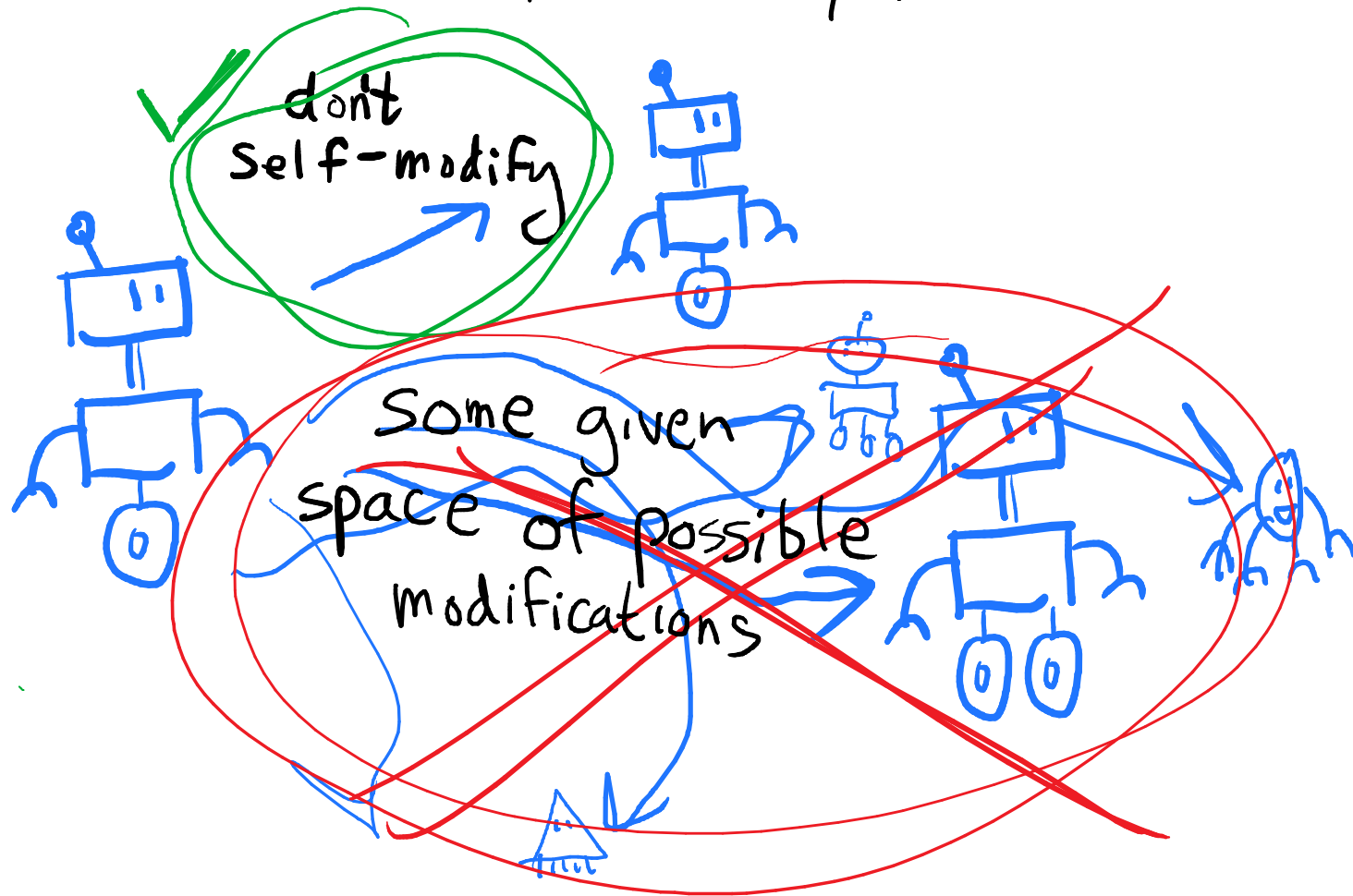
Intro to Series

- * I'm trying to communicate ideas related to my research agenda which currently exist mainly in my head and/or hard-to-digest non-public notes
- * The first big objective is to "save UDT" by finally combining it with a rigorous theory of logical uncertainty (aka. computational uncertainty, bounded rationality) as Wei Dai originally intended. That's what I'll discuss in this talk.
- * I also have ideas about moving beyond UDT, which later lectures in the series will discuss.

A property of an agent "tiles" or "is reflectively stable" if the agent would not deliberately self-modify to change that property, given the opportunity.



A "tiling theorem" (as I see it) should establish that some set of properties tile, under some given conditions, by showing that an agent with the opportunity to choose from some set of self-modifications will reject (or at least not strictly prefer) self-modification.



Vingean Tiling

- We want tiling results to be "Vingean":
the agent can trust its successor
without predicting what the successor
thinks/does in detail.
- So, tiling results are really about trust;
characterizing one agent (justifiably)
trusting another.

- # We can imagine generalizing the "trust" picture beyond static tiling, as well, although that will not be the focus here.
- # When is one agent OK with another agent's decisions?



Does Alice want to modify Bob?

- # This is known as "deference" in philosophy, & relates to the "principal-agent" problem in economics.

Four main motivations:

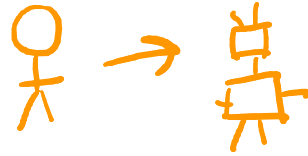
1. Human $\rightarrow$ Human
Rationality



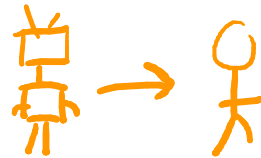
2. AI $\rightarrow$ AI
Stability



3. Human $\rightarrow$ AI
Alignment

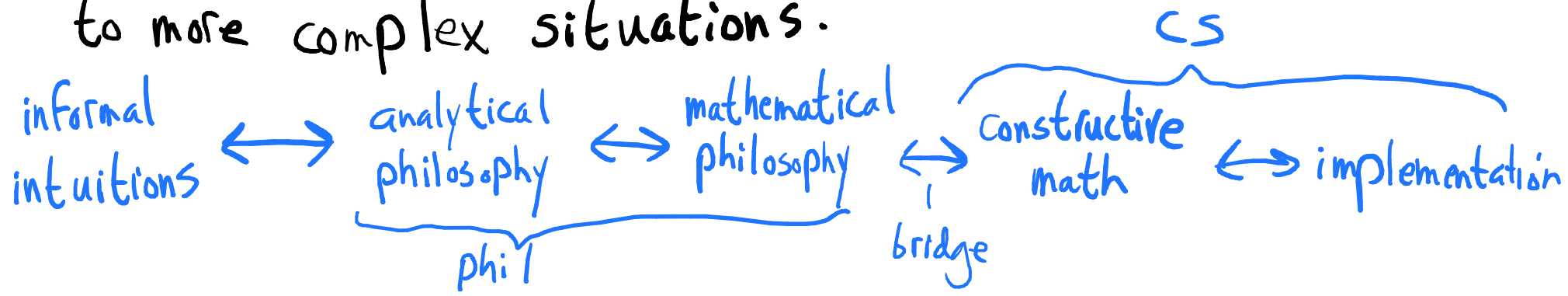


4. AI $\rightarrow$ Human
Corrigibility



How I think about formality:

- Any formal result will come with assumptions.
- Such assumptions will be unrealistic, or at least questionable.
- Think of such results as sanity checks for proposed decision procedures. We want ways of thinking which at least have good properties in simple cases.
- So a crux for the effectiveness of this sort of work is how well insights transfer from axiomatic models to more complex situations.



"Fairness" Assumptions

- I will call some of the technical assumptions "fairness" assumptions.
- This terminology is a descendant of ideas in the Timeless Decision Theory paper.
- A decision problem is "unfair" if it depends on arbitrary internal details of the agent, EG, whether the AI is written in Python or C++.
- Such problems are "unfair" because they could be used to produce "counterexamples" to any target decision theory, so, "not a fair test".

Notation

- $o \in \mathcal{O}$ observations (includes memories)
- $a \in \mathcal{A}$ actions, $\mathcal{A}_o \subset \mathcal{A}$ actions available upon seeing o
- $A(o) = a$ agent A does a upon seeing o
- $\pi \in \Pi: \mathcal{O} \rightarrow \mathcal{A}$ policies s.t. $\pi(o) \in \mathcal{A}_o$; π^* the chosen policy
- $o \mapsto a$ policy-point (as element of policy)
- $a_m \in \mathcal{A}_m \subset \mathcal{A}$ self-modifying actions, associated with a set of modifications $o \mapsto a$
- $m(\pi)$ effective policy; "runs" all the self-modifications, removing them
- $m(a)$ non-self-modifying (but outwardly identical) version of act a ; $m(a) \notin \mathcal{A}_m$. $m(a) = a$ if $a \notin \mathcal{A}_m$

*

: The chosen policy, π^ , is the input-output function associated with the agent's source code regarded mathematically. The "effective policy" $m(\pi^*)$ takes into account self-modifications.

"Running" a self-modifying action $\pi(o) = a$ in π means constructing
$$\pi'(o') = \begin{cases} m(a) & \text{if } o' = o \\ a' & \text{if } o' \mapsto a' \text{ is one of } a's \text{ modifications} \\ \pi(o') & \text{otherwise} \end{cases}$$

A self-modification can (as far as we know) only impact the future, not the past. Arbitrary webs of self-modification can create race conditions or loops.

$\Rightarrow$ So, we have to rule this out.

Assumption: there is a well-defined limit of "running" self-mod, i.e., the effective policy.

Updateless Decision Theory (UDT)

due to Wei Dai

UDT 1.0:

$$\underbrace{A(o)}_{\text{action chosen by agent upon observation } o} = \operatorname{argmax}_{\underbrace{a \in A_o}_{\text{selecting actions out of an action set}}} \mathbb{E}_{\underbrace{P_\theta}_{\text{prior}}}(\underbrace{u | A(o)=a}_{\text{condition on one action}})$$

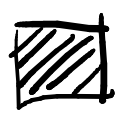
$$\text{UDT 1.1: } \underbrace{\pi^*}_{\substack{\text{agent's policy} \\ (\pi^*(o) = A(o))}} = \operatorname{argmax}_{\underbrace{\pi \in \Pi}_{\text{selecting from set of possible policies}}} \mathbb{E}_{P_\theta}(u | \underbrace{\pi^* = \pi}_{\text{condition on full policy}})$$

Note: Wei Dai specifically wants UDT to condition on the mathematical statement that the agent's source-code has such-and-such input-output behavior. A more careful statement of this requires Quines to support the recursive definitions of $A(o)$ and π^* above.

UDT 1.1 has a very simple tiling property:

Fairness Assumption:

$$\forall \pi_1, \pi_2 \quad m(\pi_1) = m(\pi_2) \Rightarrow \\ \mathbb{E}(u | \pi^* = \lceil \pi_1 \rceil) = \mathbb{E}(u | \pi^* = \lceil \pi_2 \rceil)$$

1. Suppose for contradiction $\exists \pi, o$ such that $\pi(o) \in A_m$ and $\mathbb{E}(u | \pi^* = \lceil \pi \rceil) > \mathbb{E}(u | \pi^* = \lceil \pi' \rceil)$ for all π' containing no self-modification
2. $\mathbb{E}(u | \pi^* = \lceil \pi \rceil) = \mathbb{E}(u | \pi^* = \lceil m(\pi) \rceil)$ by fairness
3. contradiction with step 1. 

This result is very limited. Because the whole policy is chosen at once, there are not really two distinct decision points such that one can trust another. This blocks us from studying Vingean trust within UDT1.1.

UDT1.0 seems more promising, though. There are multiple decision points, which need not be sure of each other's exact behaviors.

UDT 1.0 Tiling

Coordination Assumption:

$$\forall \pi, o: \mathbb{E}(u | \pi^* = \lceil \pi \rceil) \geq \mathbb{E}(u | \pi^* = \lceil \pi^* \rceil) \Rightarrow \\ \mathbb{E}(u | \pi^*(o) = \lceil \pi(o) \rceil) \geq \mathbb{E}(u | \pi^*(o) = \lceil \pi^*(o) \rceil)$$

1. $\nexists$ for $\nexists o_m: \pi^*(o_m) \in \tilde{A}_m$ and $\forall a \notin \tilde{A}_m: \mathbb{E}(u | \pi^*(o_m) = \lceil \pi^*(o_m) \rceil) > \mathbb{E}(u | \pi^*(o_m) = \lceil a \rceil)$

2. $\mathbb{E}(u | \pi^* = \lceil m(\pi^*) \rceil) = \mathbb{E}(u | \pi^* = \lceil \pi^* \rceil)$ by fairness

3. $\mathbb{E}(u | \pi^*(o_m) = \lceil m(\pi^*)(o_m) \rceil) \geq \mathbb{E}(u | \pi^*(o_m) = \lceil \pi^*(o_m) \rceil)$ by coordination

4. Contradiction. 

This is better, but still not very satisfying; Coordination plays too large a role. We need to assume away coordination problems to justify UDT1.0, but we want Vingean self-trust to follow from some understanding of the decision procedure, not a bold assumption of trustworthiness.

Some new assumptions:

Action Coordination:

$$P(\operatorname{argmax}_{a_1} \mathbb{E}(u | \pi^*(o_1) = a_1 \wedge \pi^*(o_2) = a_2) = \operatorname{argmax}_{a_1} \mathbb{E}(u | \pi^*(o_1) = a_1)) = 1 \text{ when } a_2 \notin A_m \text{ and } o_1 \neq o_2$$

Limited Self-Modification:

$a_m \in A_m$ only modify one policy-point $ob(a_m) \mapsto ac(a_m)$
with $ac(a_m) \notin A_m$ and $a_m \notin A_{ob(a_m)}$ "can't modify itself"
needed to keep fine-grained fairness from conditioning on a contradiction

Fine-Grained Fairness:

$$\mathbb{E}(u | \pi^*(o) = a_m) = \mathbb{E}(u | \pi^*(o) = m(a_m) \wedge \pi^*(ob(a_m)) = ac(a_m))$$

Faith in Argmax:

$$\forall o: \mathbb{E}(u | \phi \wedge \pi^*(o) = \operatorname{argmax}_a \mathbb{E}(u | \phi \wedge \pi^*(o) = a))$$

$$\boxed{\phi = \pi^*(o') = a' \text{ for } o' \neq o} \geq \max_a \mathbb{E}(u | \phi \wedge \pi^*(o) = a)$$

& the agent knows its own decision procedure.

UDT 1.0 Tiling (attempt #2)

1. $\$$ for $\nexists \mathbb{E}(u | \pi^*(o_m) = a_m) > \mathbb{E}(u | \pi^*(o_m) = a)$ for $\exists a_m \in A_m$
 $\forall a \notin A_m$
2. $\mathbb{E}(u | \pi^*(o_m) = a_m) = \mathbb{E}(u | \pi^*(o_m) = m(a_m) \wedge \pi^*(ob(a_m)) = ac(a_m))$ FGF
3. $\leq \max_a \mathbb{E}(u | \pi^*(o_m) = m(a_m) \wedge \pi^*(ob(a_m)) = a)$ def of max
4. $\leq \mathbb{E}(u | \pi^*(o_m) = m(a_m) \wedge \pi^*(ob(a_m)) = \arg \max_a \mathbb{E}(u | \pi^*(o_m) = m(a_m) \wedge \pi^*(ob(a_m)) = a))$ faith in argmax
5. $= \mathbb{E}(u | \pi^*(o_m) = m(a_m) \wedge \pi^*(ob(a_m)) = \arg \max_a \mathbb{E}(u | \pi^*(ob(a_m)) = a))$ restricted coordination
6. $= \mathbb{E}(u | \pi^*(o_m) = m(a_m))$ knowledge of decision procedure
7. $\nexists$ with #1. 

(Inspired by Linda Linsefors)

- * Vingean tiling.
- * Comes from reasoning about the decision procedure.
- * Reasoning is rough/weird.
- * Not clear how much sense it makes for the agent to reason about itself this way.
- * Omniscience/optimism about how good optimization will be, combined with ignorance about specific $\arg\max$

Clearly, the crux is how we handle computational uncertainty.

Radical Probabilism (Richard Jeffrey)

- # Bayesian evidence is always probability 1
- # Jeffrey wanted to handle uncertain evidence
- # Bayesian updates are less well-justified by Dutch Book than the Kolmogorov axioms
- # If we follow the Dutch Books, we only get Bayesian updates in expectation
- # This turns out to work very well for computational uncertainty

Market Analogy

- # We are used to thinking about Bayesianism in terms of Dutch Book arguments.
- # This strategy generalizes well to bounded rationality.
- # We want to construct something not easily Dutch Booked
 - low description complexity
 - low computational complexity

Logical Induction: Basic Idea (Scott G et al)



- # Specify some language $\mathcal{L}$
- # Specify a class of continuous trading strategies, mapping prices to specific trades
- # Specify a class of traders
 - ↳ e.g. poly-time as a function of t , mapping t to a trading strategy
- # Specify initial wealths for traders (eg, $2^{-(\text{description length})}$)
- # Specify schedule for traders to enter the market

LI Basic Idea Continued

Logical Induction Criterion:

- * No trader can exploit the market (unboundedly).
(Basically, money-pump the market.)

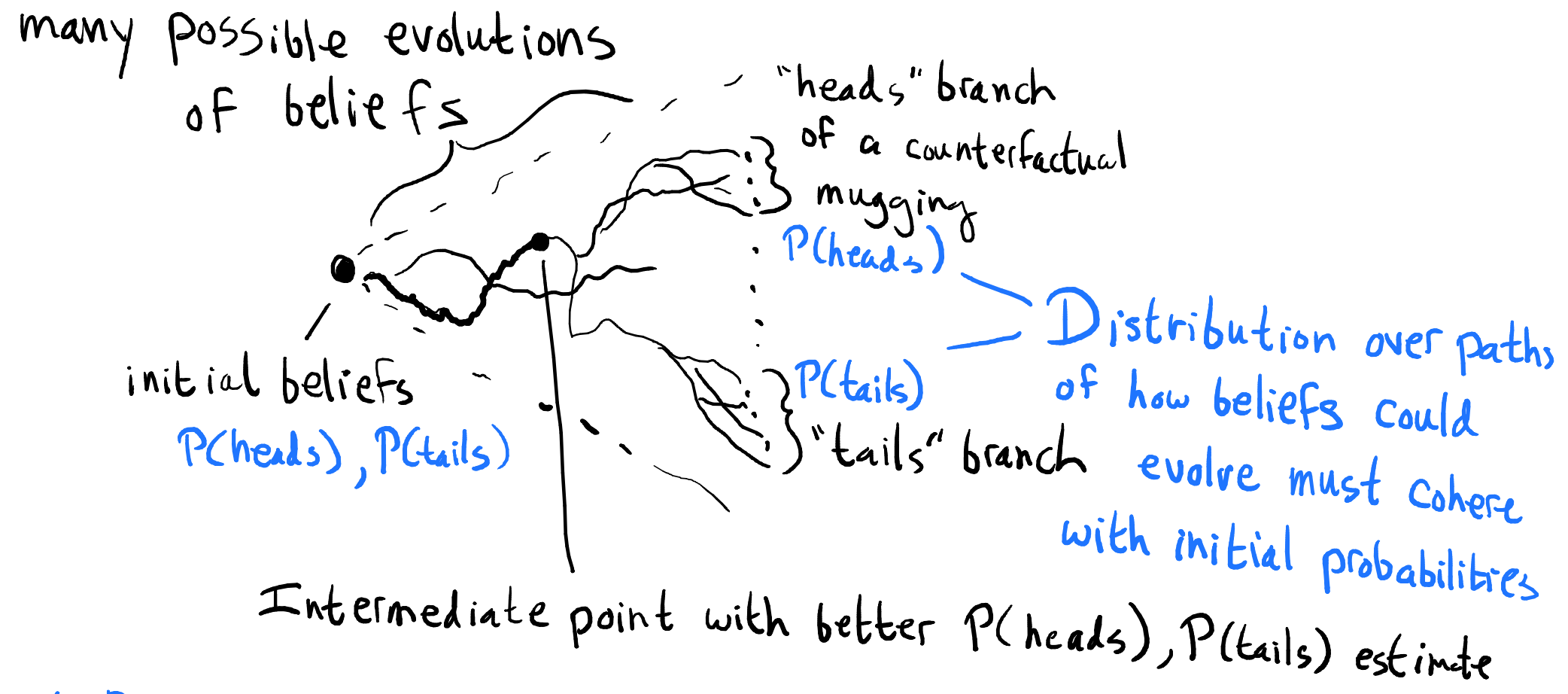
Logical Induction Algorithm:

- * At each time t , the "house" sets prices so as to balance traders against each other, based on their wealths. This is possible via Brouwer fixed-point.
- * Trader wealths at t are updated from $t-1$ based on the new market prices (which are themselves based on wealths), so wealths & prices are one big fixed-point.

First Problem: Terrible Prior

- # An early market state will assign very poor estimates to almost all market goods.
- # Therefore, it will make terrible decisions.
- # UDT locks those things in forever.
- # So UDT seems terrible.
- # My hope is to spell out extra rationality conditions sufficient to avoid this problem.
 - ↳ However, I will come back to the "terrible prior" problem in a later talk in this series.

What Computational Updatelessness Should Be



UDT should choose an intermediate point to make its decisions from, by virtue of the usual reasons why UDT can behave updatefully.

Counterfactual Mugging with Evidence

- # Motivating analogy: UDT should handle computational uncertainty much like it handles empirical uncertainty.
- # Imagine Omega selects a coin bias uniformly, but then, shows you several flips before cf. mugging you.

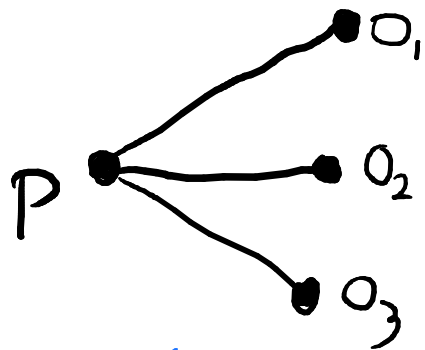
Version 1: Omega gives you money proportional to the total prob. you give Omega money

Version 2: Omega gives you money proportional to prob. you deliver given the evidence you saw

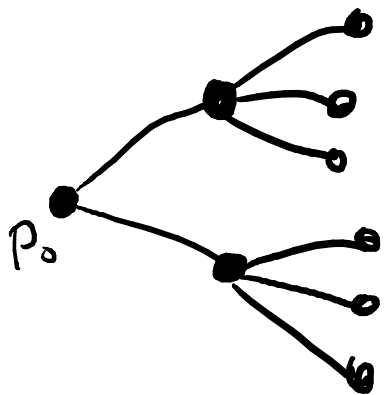
UDT behaves updatefully unless it has reason not to:

UDT's calculations decompose across branches:

$$\mathbb{E}_p(u|\phi) = \sum_{o=o_1, o_2, o_3} P(o|\phi) \mathbb{E}_p(u|\phi, o)$$



So if a policy-point choice $o_1 \mapsto a_1$ vs $o_1 \mapsto a_2$ *(significantly)* does not impact other branches ($\mathbb{E}_p(u|o_1 \mapsto a, o_{\neq 1}) = \mathbb{E}_p(u|o_{\neq 1})$) and does not impact branch probabilities ($P(o|o_1 \mapsto a) = P(o)$), then UDT will behave in an updateful manner.



Similarly, in cases where UDT can perfectly recall a sequence of observations, we can recursively decompose, & conclude UDT will behave "as updatefully as it can"

How can we constrain LI to behave like this?

Prop 1

y	n
---	---

prop 2

y	n
---	---

prop N

y	n
---	---

Propositional Coherence

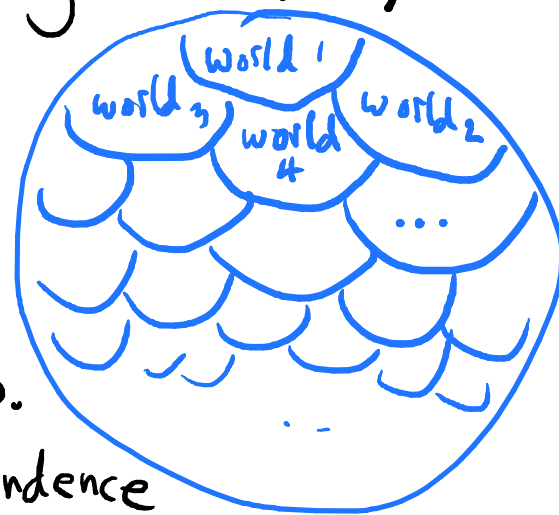
* One way to set up LI is to imagine one "betting table" per proposition. Traders put money on "yes" or "no" indicating their bets, & the house's job is to set odds so that the ratio of money set down = the odds.

- The market will converge towards respecting propositional coherence rules $P(T)=1$, $A \wedge B = 1 \Rightarrow P(A \vee B) = P(A) + P(B)$ because the house rewards arbitrage of failures of these rules.

- Alternatively, we can enforce prop. coh. right away by instead setting up one big betting table on all 2^N boolean combinations.

- Traders bet on a single prop. by spreading money out.

- This only enforces coherence for finitely many prop. at once, but we can set the rest naively assuming independence



Propositional coherence ensures that conditional probabilities work out as desired, but this alone does not give us the decomposition we wanted:

$$\mathbb{E}_P(u|\phi) = \sum_{o=o_1, o_2, o_3} P(o|\phi) \mathbb{E}_P(u|\phi, o)$$

We need an analogous coherence for expectations.

We could try several things, including an analogous "big table" approach which incorporates constraints about real-valued variables rather than only Booleans.

I'll sketch an approach based on the Jeffrey-Bolker axioms:

Jeffrey-Bolker preference structures $\mathcal{J}$

* Event algebra is a Boolean algebra. $\vee, \wedge, \neg, \top, \perp$

* $P(E)$ is a probability distribution obeying the usual axioms minus countable additivity.

$$\Leftrightarrow P(\top) = 1, A \wedge B = \perp \Rightarrow P(A \vee B) = P(A) + P(B)$$

* $V(E)$ acts like an expected utility:

$$V(A \vee B) = \frac{P(A)V(A) + P(B)V(B)}{P(A) + P(B)} \text{ if } A \wedge B = \perp$$

($V(\perp)$ is not defined)

* Equivalently, $\mathcal{J}$ is a finitely-additive vector-valued measure

with vectors $\mathcal{J}(E) = [P(E), V(E) \cdot P(E)]$

$$\Leftrightarrow \mathcal{J}(A \vee B) = \mathcal{J}(A) + \mathcal{J}(B) \text{ when } A \wedge B = \perp$$

(Based on Scott G's "Multigrain Models")

- A "trading strategy" or high-level belief h has type $\overbrace{j \rightarrow j}^{(\text{continuous})}$
 - $\hookrightarrow$ Can define weighted mixtures $h_3(x) = a h_1(x) + b h_2(x)$
 - $\hookrightarrow$ Collapse to j by taking a fixed point
 - $\hookrightarrow$ Ordinary hypotheses are just constant functions.
 - $\hookrightarrow$ Can encode constraints; mixtures will take intersection of compatible constraints, or try to softly violate incompat. constraints.

-- We still want computability, so h should be finitely-specified, with some simple default behavior for E not mentioned, like how we assumed independence to extend propositional coh. to infinitely many sentences.

$\hookrightarrow$ Would be interesting to take a more principled approach

Logical Induction Adapted to JB

- # A trader t_i emits a sequence of trading strategies $t_i(1), t_i(2), t_i(3), \dots$ and has initial weight ("wealth") w_i^0
- # We imagine traders Kelly-bet on stocks $V(x)$ and $P(x)$
- # "Stock Prices" $V_d(x), P_d(x)$ are given by a chosen fixed point of $\mathbb{I}_d = \sum_i w_i^d \cdot t_i(d)$
- # w_i^d depends on the value of investments $t_i(d-1)$ as evaluated at d (so w_i^d are part of the fixed point)
- # P_d is a prediction market; V_d is an impact market

Second Problem: What Are the Observations?

Again, we want UDT's calculations to decompose:

$$\mathbb{E}_p(u|\phi) = \sum_{o \in \text{some mutually exclusive \& jointly exhaustive set}} P(o|\phi) \mathbb{E}_p(u|\phi, o)$$

- # Logical Induction doesn't come with an obvious notion of "observation" to represent the "information" it gets by thinking longer, & Radical Probabilism explicitly denies the idea that updates are always associated w/ observing propositions which justify the update — rational agents can change their minds
- # Yet, we need a notion of observation to construct the "branches" o in the decomposition, & also the policy information in ϕ (policy-points $o \mapsto a$ for UDT 1.0)

Bayesian Logical Induction (Sam Eisenstat)

Basic idea: model LI as observing & updating on
the whole set of prices

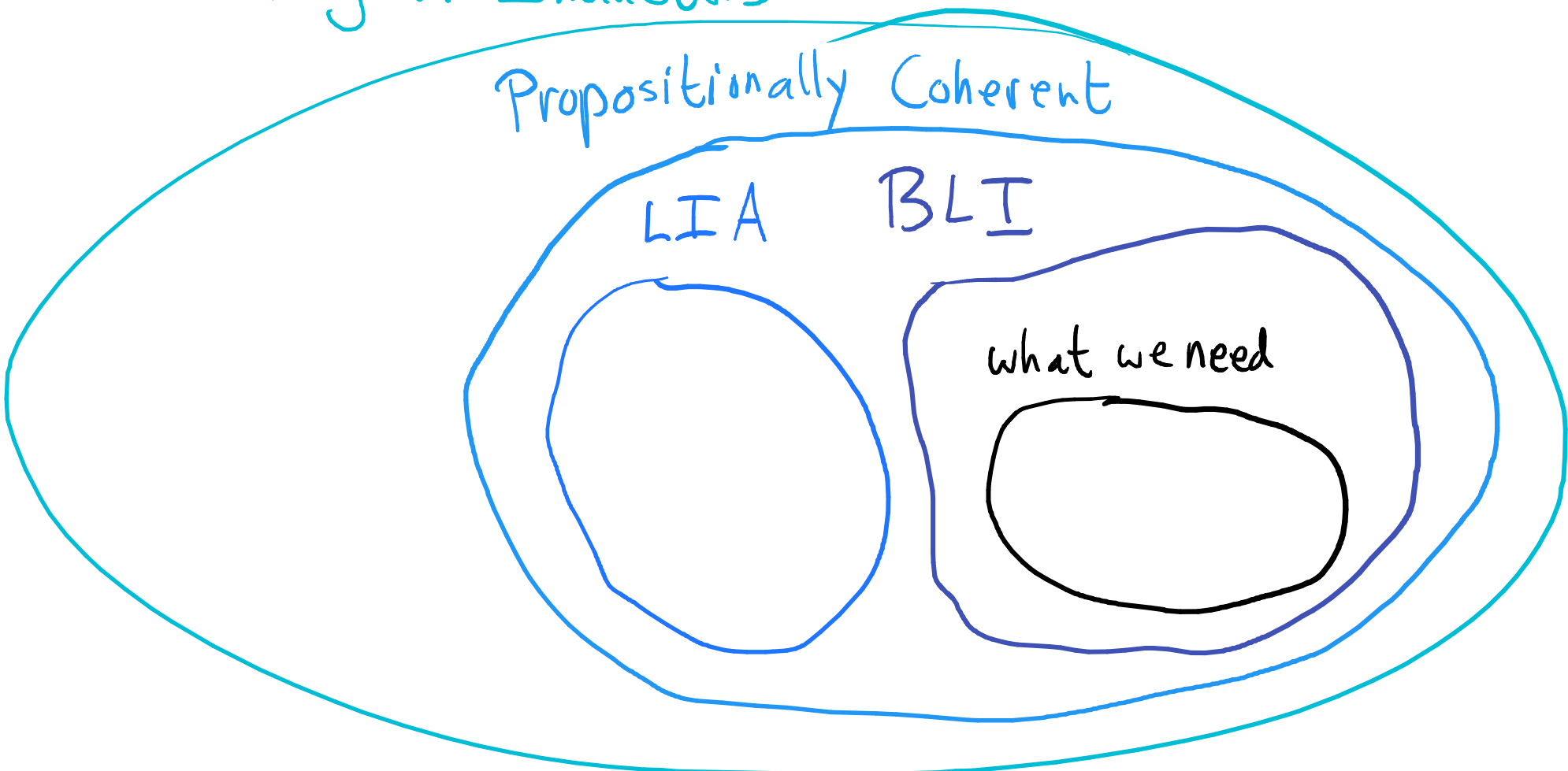
Logical Inductors

Propositionally Coherent

LIA

BLI

what we need



Bayesian LI Cont.

Philosophically: what are you "observing" if you think for some time & change your mind?

↳ You could be observing some argument or some computation, but Radical Probabilism wants to leave room for "freely" changing your mind.

↳ This might seem irrational, but it is a useful exercise to try & keep our notion of rationality broad, not adding extra assumptions unless really needed.

Therefore, "observing" beliefs themselves leaves agents with the freedom to come up with these beliefs in whatever way they like, & doesn't treat any type of evidence (eg "empirical observation") as fundamentally special, while still giving us a notion of "observation" which includes enough

Bayesian LI Cont.

Sam's approach is to take a propositionally coherent logical inductor $Q_1, Q_2, Q_3 \dots$ and transform it to a Bayesian logical inductor $P_1, P_2, P_3 \dots$ which meets additional rationality constraints.

The transformation modifies only the prices of "large" sentences, so that arbitrary price changes don't threaten the LIC.

Bayesian LI Cont.

I will use Q_n to denote some notation (within $\mathcal{L}$) for a list of prices for "small" sentences on day n (ie, a list of prices Q_n could have).

Sam's constraints: Approximation due to some discretization

1. $P_n(\phi) \approx Q_n(\phi)$ for ϕ which are small on day n
2. $P_n(\phi | Q_m = Q_m) := Q[n]$ when $m > n$
3. $P_n(\phi | Q_m = Q_m \wedge Q_{m+1} = Q_{m+1}) = Q_{m+1}$; similarly for longer conjunctions
4. Beliefs $P_{n-1}(Q_n = Q_n)$ are required to balance the equation

Summing over the finite discretization $P_{n-1}(\phi) = \sum_{Q_n} P_{n-1}(Q_n = Q_n) Q[\phi]$

Bayesian LI Cont.

to within vanishing ϵ

$\mathbb{P}$ is now a logical inductor, since it agrees with $\mathbb{Q}$ on small prices; furthermore, we can see $\mathbb{P}$'s updates from $\mathbb{P}_n$ to $\mathbb{P}_{n+1}$ as a Bayesian update, since

$$\mathbb{P}_{n+1}(\phi) = \mathbb{P}_n(\phi | Q_{n+1} = Q_{n+1})$$

where Q_{n+1} denotes Q_{n+1} 's actual prices.

Notice that Sam's constraints 1 & 4 imply:

$$\mathbb{P}_{n-1}(\phi) = \sum_{Q_n} \mathbb{P}_{n-1}(Q_n = Q_n) \mathbb{P}_{n-1}(\phi | Q_n = Q_n)$$

This is a version of conservation of expected evidence, AKA no expected net update.

Bayesian LI Cont.

- + Just listing the constraints doesn't prove that a BLI exists.
- + Constraints 1-3 are easy enough (we can modify large sentences however we like).
- + We can show that #4 is feasible by choosing $P(Q_n = Q_n) = 1$ for one specific Q_n such that $Q_{n-1}(\phi) = Q_n(\phi)$, and $P(Q_n = Q_n) = 0$ otherwise.
- + Obviously this is a very unsatisfying choice!
- + This illustrates how BLI is a very incomplete set of "rationality constraints" for large sentences.

Deriving BLI More Naturally?

- # Just as Propositional Coherence precisely enforces rationality constraints which any LI learns in the limit, we would like to derive BLI by precisely enforcing rationality constraints which LI provably learns.
- # A natural idea is to universally enforce

$$P_n(\phi | P_m(\phi) = p) = p$$

However, this condition is inconsistent with some self-referential sentences, such as

$$L := P_m(L) < \frac{1}{2}$$

- # The weaker property which LI actually learns in the limit is:

$$\mathbb{E}_n(\mathbb{I}(\phi) \cdot \text{Ind}_\delta(P_m(\phi) \geq p)) \geq p \cdot \mathbb{E}_n(\text{Ind}_\delta(P_m(\phi) \geq p))$$

where $m > n$, $\mathbb{I}(\phi) = \begin{cases} 1 & \text{if } \phi \\ 0 & \text{if } \neg \phi \end{cases}$, $\text{Ind}_\delta(x \geq y) = \begin{cases} 0 & \text{if } x \leq y \\ \frac{x-y}{\delta} & \text{if } y < x \leq y+\delta \\ 1 & \text{if } y+\delta < x \end{cases}$

So, can this be enforced exactly? What does it get us?

Third Problem: Unpacking Obs (Work with Martín Soto)

→ BLI (seemingly) allows us to sensibly extrapolate "small" market beliefs about what may happen in the future to a "large" distribution over all possible future market states, respecting the desired decomposition of expectations; but, this is only adequate for a version of UDT with decision rule

$$A = \operatorname{argmax}_a \nabla_{\theta} (O=o \Rightarrow A=a)$$

which is a terrible version of UDT.

→ We want $A(o) = \operatorname{argmax}_a \nabla_{\theta} (A(o)=a)$, but conditioning on $A(o)=a$ doesn't have nice coherence properties (yet)

Unpacking Obs

(Work with Martín Soto)

Note: I don't really like either of these assumptions

— Universal Instantiation: $\forall x. \phi(x) \rightarrow \phi(c)$

— Hypothetical Marginalization:

↳ $v(Q, x)$, $p(Q, x)$ pull out the V , P values from a large epistemic state Q

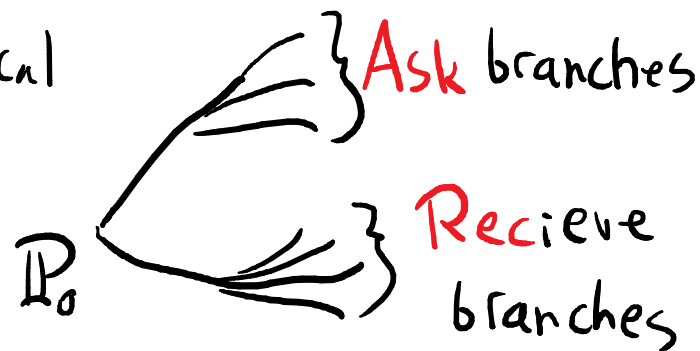
↳ We require the prior P_0 just knows the output to these functions

$$v(Q, x) = r \Rightarrow P_0(v(Q, x) = r) = 1$$

↳ If these long descriptions Q are just conjunctions of claims about real variables, this probably follows from real coherence

Unpacking Obs (cont.)

Counterlogical
Mugging:



Viability Sketch:

$\nV P_0$ thinks P_{10} will satisfy either Ask or Rec with 98% probability (49% each), and:

$$\forall Q' : \text{Ask}(Q') \rightarrow \left[[A(Q') = \text{give} \rightarrow u = -10] \wedge [A(Q') = \text{refuse} \rightarrow u = 0] \right]$$

$$\forall Q' : \text{Rec}(Q') \rightarrow$$

$$\left[\left[\forall Q'' : \text{Ask}(Q'') \rightarrow [A(Q'') = \text{give}] \rightarrow u = +100 \right] \wedge \right.$$

$$\left. \left[\neg \left[\forall Q'' : \text{Ask}(Q'') \rightarrow [A(Q'') = \text{give}] \rightarrow u = 0 \right] \right] \right]$$

Unpacking Obs (cont.)

Viability Sketch cont.:

So the question is whether those four cases evaluate to the #s we want

$A(a') = \text{give}$ policy $A(a') = \neg \text{give}$ policy

Ask branch

Rec branch

-10	0
+100	0

So long as Ask and Rec are definable from v and p , Universal Instantiation plus Hypothetical Marginalization makes this work out!

Saving UDT

- So now, UDT 1.0 for computational uncertainty works just like empirical UDT 1.0
- It behaves updatefully w.r.t. any info it thinks it should
- It obeys simple tiling theorems if we assume away self-coordination problems (which we need UDT 1.1 to solve).
 - ↳ I refer to the two theorems about UDT 1.0 from earlier; but I want a better than leveraging Lt

Conclusion

- # I've given some very basic tiling results.
- # Obviously, there is a long way to go before we can apply such results to get helpful insights for the human-to-AI case.
- # Big obstacle is the self-coordination assumption; this seems unfounded without UDT1.1, but UDT1.1 blocks interesting tiling results. How can we do better?

- overall shape of "alignment"?
- general shape of tiling agenda (?)
 - bayes obstacle $\rightarrow$ ref. oracles $\rightarrow$ comp. uncertainty $\rightarrow$ val. unc. $\rightarrow \dots$
- policy alignment: need to align full pref. structure, beliefs + values
- but what is "the human prior"?
 - not how we actually think!
 - not somewhere in the brain! (sig. interp. work required)
 - limit of philosophical deliberation (under good conditions)
- So how can we model this deliberation?
 - market models $\Rightarrow$ Fix DT
 - tiling theorem!

decision auction	impact market
prediction market	groundless induction